

Siqi Dai

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EDUCATION

University of Wisconsin-Madison

Bachelor of Science in Computer Science, Minor in Mathematics — GPA: 4.0/4.0

Madison, WI

Expected May 2028

Relevant Coursework: Data Structures & Programming, Object-Oriented Programming, Computer Engineering, Machine Learning, Optimization, AI-Assisted Software Dev, Human-Computer Interaction

TECHNICAL SKILLS

Programming: Java, Python, C++, JavaScript, HTML/CSS, SQL

Tools: Git/GitHub, Halo, VS Code, MeshLab

Libraries: NumPy, Pandas, Matplotlib, TensorFlow, scikit-learn, OpenCV

EXPERIENCE

Madability Lab, University of Wisconsin-Madison

Student Research Assistant – AR/VR Accessibility Systems

Madison, WI

Apr 2026 – Present

- Designing AR/VR navigation assistance systems for people with low vision, focusing on stairway detection, spatial awareness, and safe indoor mobility across diverse architectural environments.
- Prototyping real-time stair and edge detection pipelines using computer vision and Bayesian filtering techniques, targeting sub-50ms perception latency for interactive navigation feedback.
- Collaborating on iterative user-centered accessibility research workflows involving environment analysis, interface evaluation, and assistive navigation design.

Cybersecurity Operations Center

Student Analyst

Madison, WI

May 2026 – Present

- Investigate and triage phishing incidents using Cherwell ticketing workflows, including analyzing reported emails, creating incident cases, and coordinating response communication with affected departments.
- Monitor and analyze Palo Alto Cortex XSOAR playbooks involving outbound critical traffic, URL filtering, and cloud security alerts to support threat detection and incident escalation workflows.
- Perform endpoint and network investigation tasks by identifying device hostnames, MAC addresses, and associated user activity across campus-managed systems.

College of Letters & Science IT

L&S IT Service Desk Technician

Madison, WI

Jan 2026 – May 2026

- Delivered Tier-1 technical support for faculty, staff, and students across multiple L&S departments through phone, chat, email, and in-person troubleshooting.
- Resolved software, hardware, authentication, and network connectivity issues across Windows and macOS environments while documenting 30+ service cases in Halo.
- Maintained operational readiness of 3+ instructional computing labs through routine diagnostics, maintenance coordination, and technical room inspections.

PROJECTS

Self-Improving Image Editing Framework with Multimodal LLM Feedback

Advisor: Dr. Gupta

Apr 2026 – Present

- Developed a closed-loop image editing framework using multimodal LLM agents to iteratively generate, critique, and refine edits across multi-stage editing tasks.
- Built object-consistent editing pipelines preserving subject identity, scene composition, and lighting conditions while generating 100+ edited image variants from limited source inputs.
- Designed evaluation metrics for instruction alignment, foreground preservation, lighting consistency, and visual realism to quantitatively compare iterative editing quality.

Multi-Parameter Virtual Sensing Using Neural Networks for Automotive Engines

Advisor: Dr. Gupta

Oct 2025 – Feb 2026

- Designed and trained a multi-output ANN architecture (6→64→32→16→3) for simultaneous prediction of MF,IA, NO_x,EO, and SOC combustion parameters.
- Achieved 98.9% average predictive accuracy ($R^2 = 0.989$) while outperforming Random Forest, SVR, and OLS baselines across all target variables.
- Conducted permutation-based sensitivity analysis to validate model interpretability and feature influence consistency with combustion domain knowledge.

LEADERSHIP & ACTIVITIES

Central States Combustion Institute Undergraduate Research Competition – Winner

Apr – May 2026

- Presented machine-learning-based virtual sensing research to researchers and industry professionals at the Central States Combustion Conference hosted at Purdue University.
- Recognized with the undergraduate research competition award for technical presentation quality and research contribution.

Undergraduate Research Symposium – Presenter

Apr 2026

- Presented interdisciplinary ML research on virtual sensing systems to faculty and student audiences at UW-Madison.
- Research poster selected for permanent public display in College Library following symposium evaluation.

CheeseHack 24H Hackathon – Team Leader

Mar 2026

- Led a 4-member team in developing *MediMenu*, an AI-powered nutrition platform generating personalized meal recommendations based on dietary restrictions and health preferences.
- Built a full-stack application using React + Vite, React Router, vanilla CSS, FastAPI, and SQLite to support interactive user workflows and lightweight backend data management.
- Integrated OpenAI GPT-4o Vision and structured JSON outputs to analyze food-related inputs and generate context-aware dietary recommendations.

Badger Build Festival 24H Hackathon – Team Leader

Nov 2025

- Led development of *CourseOptimizer*, a schedule optimization platform for course planning and conflict reduction.
- Implemented responsive front-end components using HTML/CSS/JavaScript and coordinated version control workflows through GitHub.

HONORS & AWARDS

- Scholarship: Mary Schreiber LeBlanc & Duwayne LeBlanc Scholarship, UW-Madison Undergraduate Scholarship for Summer Study
- Certificate: Summer 2026 Poster Display at College Library Certificate of Recognition
- Deal List: 2026, 2025 L&S Dean's List